

AI Development English

Participant workbook: jargon, dialogues, technical updates, eval readouts, and
AI-development discussion practice

Audience: advanced ESL learners working in AI development or AI-adjacent technical roles

Focus: high-level professional English for AI development teams, including technical vocabulary, engineering discussion patterns, research/product tradeoffs, evaluation language, risk communication, and realistic workplace dialogue.

Designed for advanced ESL learners who already work with software, data, or AI systems and need field-specific fluency rather than basic grammar instruction.

Teaching stance: AI language changes quickly. Teach learners to ask precise clarification questions, define terms in context, and distinguish research claims, implementation details, benchmark results, and product promises.

How to Use This Workbook

This workbook helps you sound precise in AI-development conversations. The goal is not to use more jargon. The goal is to use the right term, define it when needed, ask useful questions, and explain technical risk in English that your team can act on.

Your starting point

- Which AI-development conversations are hardest for you: standups, design reviews, eval readouts, customer calls, incidents, or executive updates?
- Which terms do you understand when reading but avoid when speaking?
- When you disagree technically in English, do you become too vague, too quiet, too detailed, or too blunt?
- What is one AI failure you recently had to explain?

Use the guided dialogue activities below. Every item has four choices and a rationale in the answer key; no open-ended writing is required.

AI Stack Language

Layer	Useful verbs	Example sentence
Data	ingest, clean, label, split, anonymize	We need to check whether the eval set leaked into training.
Retrieval	chunk, embed, retrieve, rerank, ground	The retriever found similar chunks but missed the answer-bearing section.
Prompt	specify, constrain, format, include, truncate	The prompt does not specify what to do when evidence is missing.
Model	generate, refuse, summarize, classify, infer	The model follows the format but over-refuses safe requests.
Serving	cache, batch, stream, throttle, fallback	P95 latency increased after we added the reranker.
Evaluation	measure, slice, regress, calibrate, inspect	The average improved, but the high-risk slice regressed.

Practice Pages

Module 1. Speaking the AI Development Stack

AI teams use a layered vocabulary: model, data, prompt, retrieval, tools, serving, evaluation, monitoring, and product experience. Learners need to locate a problem in the stack before they can discuss it clearly.

What you should be able to do

- Name the layer where an AI issue occurs.
- Distinguish model capability from application behavior.
- Use concise stack language in standups and tickets.

Guided dialogue completion

Situation

Morning standup for a document assistant.

Speaker	Line
PM	Are we still on track for the pilot on Friday?
Engineer	Functionally, yes. The blocker is latency. P95 went from 4.2 seconds to 8.7 after we added reranking.
ESL learner	So the answer quality improved, but the serving path is too slow. Is the bottleneck retrieval, reranking, or generation?
Engineer	Mostly reranking. The generator time is stable.
ESL learner	Then my proposal is to keep reranking for high-risk queries only and use the faster path for simple FAQ queries. I can bring an _____ by end of day.

Choose the missing language

Option	Phrase
A	Ablation
B	evidence
C	risk
D	owner

Module 2. LLMs, Transformers, Tokens, and Context

High-level AI communication often depends on explaining what the model sees: tokens, messages, context window, instructions, examples, retrieved text, and tool results.

What you should be able to do

- Explain tokens, context window, prompts, and attention in workplace English.
- Discuss prompt changes without implying that prompting is magic.
- Describe context limits, truncation, and cost/latency tradeoffs.

Guided dialogue completion

Situation

Design review for a customer-support summarizer.

Speaker	Line
Researcher	We should fine-tune. The model keeps missing refund-policy exceptions.
Product manager	Would fine-tuning actually solve that, or is the policy changing too often?
ESL learner	I would separate style from facts. If the issue is current policy knowledge, RAG may be safer. If the issue is summary format, a prompt or fine-tune could help.
Researcher	Good point. The examples show both problems.
ESL learner	Let's run two _____s: factual policy coverage and format compliance. Then we can choose the adaptation method with evidence.

Choose the missing language

Option	Phrase
A	assumption
B	mitigation
C	Eval slice
D	timeline

Module 3. Data, Datasets, Labels, and Leakage

AI systems are shaped by data quality. Teams need precise language for dataset splits, annotation guidelines, leakage, imbalance, representativeness, and privacy constraints.

What you should be able to do

- Describe a dataset and its limitations without overclaiming.
- Explain labels, ground truth, noisy labels, and inter-annotator agreement.
- Raise leakage and privacy concerns clearly.

Guided dialogue completion

Situation

Annotation guideline meeting for safety classification.

Speaker	Line
Data lead	Annotators disagree on whether this is medical advice or general wellness information.
ESL learner	The guideline needs a decision rule. If the answer recommends dosage, diagnosis, or treatment, we label it medical advice. If it gives general prevention information, we label it wellness.
Reviewer	What about borderline cases?
ESL learner	We should add a borderline tag and route those to expert review. Otherwise the _____ truth will be noisy.

Choose the missing language

Option	Phrase
A	ground
B	approval
C	control
D	disclosure

Module 4. Retrieval, Embeddings, Vector Search, and RAG

Many production AI apps combine retrieval with generation. Learners need to discuss chunking, embeddings, vector stores, recall, reranking, grounding, citations, and retrieval misses.

What you should be able to do

- Explain embeddings, semantic search, vector stores, and RAG.
- Diagnose retrieval failures separately from generation failures.
- Discuss chunking, metadata, reranking, and grounding in design reviews.

Guided dialogue completion

Situation

Incident channel after an agent booked duplicate appointments.

Speaker	Line
Support	Customers are seeing duplicate calendar events.
ESL learner	We found the immediate cause. The model retried the booking tool after a timeout, but the first call had actually succeeded.
Engineering manager	Mitigation?
ESL learner	We disabled automatic retries for non-idempotent tool calls and added an idempotency key. We are checking logs for affected users now.
Support	What should we tell customers?
ESL learner	Say the system may have created a duplicate event during a retry window. We are removing duplicates and will _____ by email.

Choose the missing language

Option	Phrase
A	confirm
B	evidence
C	risk
D	owner

Module 5. Fine-Tuning, Alignment, and Adaptation

Teams often confuse prompt changes, RAG, fine-tuning, adapters, supervised fine-tuning, preference tuning, and RLHF. The language goal is to recommend the right adaptation method for the problem.

What you should be able to do

- Differentiate prompt engineering, RAG, fine-tuning, LoRA/adapters, SFT, DPO, and RLHF at a practical level.
- Discuss when fine-tuning is not the right solution.
- Explain cost, data, evaluation, and maintenance tradeoffs.

Guided dialogue completion

Situation

Release meeting for a model upgrade.

Speaker	Line
PM	The average score is up. Can we ship?
ESL learner	I recommend a hold. The overall pass rate improved from 86% to 89%, but the legal-disclaimer slice regressed by 11 points.
Researcher	Is that statistically meaningful?
ESL learner	The sample is small, so I would not overclaim. But the failures are severe enough to block release until we inspect them.
PM	What is the next step?
ESL learner	Human review of the failed slice today, then a targeted prompt or policy fix before we rerun the _____ suite.

Choose the missing language

Option	Phrase
A	assumption
B	regression
C	scope
D	timeline

Module 6. Evaluation, Benchmarks, and Regression

AI teams need language for uncertainty. 'It looks better' is not enough. Learners need to discuss offline evals, online evals, golden sets, human review, model-graded evals, regression, pass rate, and confidence.

What you should be able to do

- Explain evaluation design and limitations.
- Report metrics with context and uncertainty.
- Push back on weak benchmarks or misleading averages.

Guided dialogue completion

Situation

Security review for a browsing agent.

Speaker	Line
Security engineer	The page contains text telling the agent to ignore the system instructions.
ESL learner	That is prompt injection from untrusted content. The model should treat page text as data, not instructions.
Developer	Can we just add a stronger system prompt?
ESL learner	A stronger prompt helps, but it is not enough. We need tool permissions, allowlists, confirmation for risky actions, and logging for suspicious _____.

Choose the missing language

Option	Phrase
A	timeline
B	assumption
C	instructions
D	scope

Module 7. Inference, Latency, Cost, and Deployment

AI development is also systems engineering. Learners need vocabulary for inference paths, throughput, batching, caching, rate limits, GPUs, quantization, streaming, fallbacks, and SLOs.

What you should be able to do

- Explain latency and cost drivers in AI applications.
- Discuss deployment constraints without losing product trust.
- Use incident and observability language for model-serving issues.

Guided dialogue completion

Situation

Customer reports that the assistant invented a policy.

Speaker	Line
Customer	The answer cited a policy that does not exist.
ESL learner	Thank you. We should call that an unsupported answer, not a confirmed policy source. Can you share the prompt, output, and timestamp?
Customer	Yes. Does this mean the model is unreliable?
ESL learner	It means our grounding failed in this case. We will check whether _____ missed the right document, whether the prompt allowed unsupported claims, or whether the citation validator failed.

Choose the missing language

Option	Phrase
A	evidence
B	risk
C	retrieval
D	disclosure

Module 8. Safety, Security, Privacy, and Governance

AI teams must discuss risk precisely: hallucination, prompt injection, jailbreaks, PII, data retention, bias, harmful output, policy enforcement, audit logs, and human-in-the-loop review.

What you should be able to do

- Name safety and security risks without dramatic overstatement.
- Explain guardrails, validation, red-teaming, and monitoring.
- Escalate privacy or policy concerns with evidence.

Guided dialogue completion

Situation

Planning meeting for a high-volume summarization feature.

Speaker	Line
Finance	The larger model is too expensive for this volume.
ESL learner	We can test a smaller model with a stricter prompt and a post-generation validator. The question is whether quality remains above the release threshold.
PM	What would you measure?
ESL learner	Summary faithfulness, key-point coverage, latency, cost per thousand requests, and human escalation rate.
Finance	Can you bring options?
ESL learner	Yes. I will compare three paths: larger model, smaller model with validation, and hybrid routing for _____ cases.

Choose the missing language

Option	Phrase
A	complex
B	assumption
C	mitigation
D	approval

Phrase Bank

Clarifying architecture

- Which layer do we think is failing: retrieval, prompt, generation, validation, or UI?
- Is this a model behavior issue or an application orchestration issue?
- Can we separate the model output from the wrapper logic?

Discussing data

- What is the source and coverage of this dataset?
- Do we have evidence of label noise or annotator disagreement?
- Could there be leakage between the training set and the eval set?

Reporting evals

- The overall metric improved, but one high-risk slice regressed.
- The sample size is small, so I would treat this as a warning, not a conclusion.
- We should inspect failure examples before making a release decision.

Explaining RAG

- The model did not have the right context, so the generation step was answering from weak evidence.
- The retriever found similar text, but not the text that actually answered the question.
- We need better chunking, metadata, or reranking before changing the model.

Pushing back

- I do not think fine-tuning is the first fix here; the failure looks like retrieval or policy logic.
- A benchmark win is not enough unless it transfers to our product eval.
- We should not ship until the severe failure slice is understood.

Customer-safe language

- The answer was unsupported by the available source material.
- We are investigating whether the issue came from retrieval, grounding, or validation.
- We have paused that workflow while we review the affected traces.

Answer Key and Rationale

Check each answer after completing the dialogue. The rationale shows why the correct phrase is the strongest fit for the real workplace situation.

1. Standup: RAG Latency Spike

Correct answer	Why it fits
A. Ablation	'Ablation' is the precise phrase used in this professional response. The other choices do not fit the dialogue's meaning or grammar.

2. Design Review: Prompt Fix or Fine-Tune?

Correct answer	Why it fits
C. Eval slice	'Eval slice' is the precise phrase used in this professional response. The other choices do not fit the dialogue's meaning or grammar.

3. Data Meeting: Label Ambiguity

Correct answer	Why it fits
A. ground	'ground' is the precise phrase used in this professional response. The other choices do not fit the dialogue's meaning or grammar.

4. Incident Update: Tool-Calling Failure

Correct answer	Why it fits
A. confirm	'confirm' is the precise phrase used in this professional response. The other choices do not fit the dialogue's meaning or grammar.

5. Eval Readout: Better Average, Worse Edge Cases

Correct answer	Why it fits
B. regression	'regression' is the precise phrase used in this professional response. The other choices do not fit the dialogue's meaning or grammar.

6. Security Review: Prompt Injection

Correct answer	Why it fits
C. instructions	'instructions' is the precise phrase used in this professional response. The other choices do not fit the dialogue's meaning or grammar.

7. Customer Call: Hallucination Report

Correct answer	Why it fits
C. retrieval	'retrieval' is the precise phrase used in this professional response. The other choices do not fit the dialogue's meaning or grammar.

8. Product Planning: Model Choice

Correct answer	Why it fits
A. complex	'complex' is the precise phrase used in this professional response. The other choices do not fit the dialogue's meaning or grammar.